EXECUTIVE BUSINESS REPORT

Revenue Optimization & Dynamic Cancellation Control

Hotel Booking Analytics & Strategic Decision Framework

BASELINE CANCELLATION RATE	PRIMARY VULNERABILITY	HIGHEST IMPACT STAYS
37.04%	Portugal (PRT)	Weekdays (4-5 Nights)

1. BUSINESS PROBLEM

The hotel group faces a significant operational and financial challenge: a **37.04% baseline cancellation rate** across both Resort and City Hotels. This causes severe revenue leakages, high overhead in empty room maintenance, and staffing inefficiencies.

Key bottlenecks identified:

- **High Geographic Concentration:** Portugal (PRT) dominates total booking volumes but possesses an alarmingly disproportionate share of cancellations.
- **Inventory Perishability:** Cancellations of long-stay bookings (particularly weekday stays) occur late in the cycle, leaving un-booked gaps that cannot easily be backfilled.
- **Lead-Time Vulnerability:** The longer a reservation sits in the system, the more likely it is to be canceled.

2. ASSUMPTIONS

1. **Price & Commitment Elasticity:** High cancellations in PRT and during peak months can be suppressed by introducing dynamic, non-refundable discount thresholds (e.g., offering lower rates in exchange for immediate financial commitment).
2. **Stricter Friction Policies on Weekdays:** Since weekday stays are longer and costlier to lose, they require tighter cancellation windows compared to brief weekend stays.
3. **Demographic Tailoring:** Segmenting guests (Couples, Singles, Families) will allow us to target high-value, lower-risk customer types with targeted marketing.

3. RESEARCH QUESTIONS

1. What is the cancellation profile of our most dominant market (Portugal), and how does it compare to international travelers?
2. How do stay patterns (weekdays vs. weekends) and guest categories dictate reservation commitments?
3. Which specific calendar months present the highest cancellations, and how can we design targeted discount schedules to stabilize booking velocity?

PYTHON CODE SNIPPET — DATA PREPARATION & CHANNEL SEGMENTATION — PANDAS / PLOTLY

```
import pandas as pd
import plotly.express as px
import plotly.graph_objects as go

# Prepare analysis variables
```

3. RESEARCH & DEMOGRAPHIC LOGIC (CONT.)

PYTHON CODE SNIPPET — DEMOGRAPHIC CATEGORIZATION & HEATMAP MATRIX CONSTRUCTION VISUALIZATION SCRIPT

```
def get_demographic_category(row):
    adults = row['adults']
    children = row['children'] if not pd.isna(row['children']) else 0
    babies = row['babies'] if not pd.isna(row['babies']) else 0

    if children > 0 or babies > 0:
        return 'Family'
    elif adults == 1:
        return 'Single'
    elif adults == 2:
        return 'Couple'
    else:
        return 'Group'

df_report['Category'] = df_report.apply(get_demographic_category, axis=1)

# Visualization 1: Dynamic Heatmap of Bookings by Country, Channel, and Demographics
df_ok = df_report[df_report['is_canceled'] == 0]
top_countries = df_ok['country'].value_counts().nlargest(10).index.tolist()
df_top_countries = df_ok[df_ok['country'].isin(top_countries)]
country_matrix = df_top_countries.groupby(['country', 'Category']).size().unstack(fill_value
=0)

fig_heat = px.imshow(
    country_matrix,
    labels=dict(x="Guest Category", y="Country", color="Successful Bookings"),
    x=country_matrix.columns,
    y=country_matrix.index,
    color_continuous_scale='Viridis',
    title='Successful Bookings Concentration Matrix (Top 10 Countries)'
)
fig_heat.show()
```

Successful Bookings Concentration Matrix (Top 10 Countries)



4. ANALYTICS & INSIGHTS

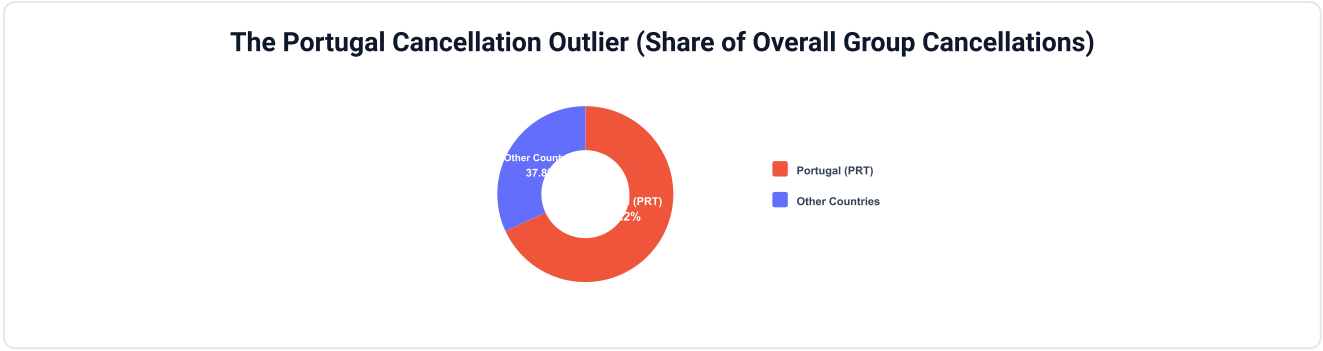
A. The Portuguese (PRT) Market Dilemma

Portugal represents our main pipeline of guests, but it is simultaneously our greatest source of volatility. As shown in the cancellation breakdown below, PRT represents the majority of our entire group's cancellation footprint, signaling that local guests use flexible reservations as zero-risk placeholders.

PYTHON CODE SNIPPET — PORTUGAL CANCELLATION OUTLIER PIE CHART — PLOTLY EXPRESS

```python
# Visualization 2: Focus on the Portugal (PRT) Outlier vs Global Markets
df_canc = df_report[df_report['is_canceled'] == 1].copy()
prt_vs_others = df_canc['country'].apply(
    lambda x: 'Portugal (PRT)' if x == 'PRT' else 'Other Countries'
)
prt_dist = prt_vs_others.value_counts().reset_index(name='cancellations')

fig_prt = px.pie(
    prt_dist,
    values='cancellations',
    names='country',
    title='The Portugal Cancellation Outlier (Share of Overall Group Cancellations)',
    color_discrete_sequence=['#ef553b', '#636efa'],
    hole=0.4
)
fig_prt.update_traces(textposition='inside', textinfo='percent+label')
fig_prt.show()
```



4. ANALYTICS & INSIGHTS (CONT.)

B. Weekend vs. Weeknight Stays Analysis

Canceled bookings show a significantly longer duration on weeknights compared to weekends. When a customer cancels a 4-to-5-night weekday stay, the hotel suffers heavily as mid-week empty slots are exceptionally challenging to fill dynamically.

PYTHON CODE SNIPPET — MONTHLY CANCELLATION VOLATILITY ANALYSIS — PLOTLY GRAPH OBJECTS

```python
# Visualization 3: Monthly Cancellation Volatility & Nights Lost
month_map = {
    'January': 1, 'February': 2, 'March': 3, 'April': 4,
    'May': 5, 'June': 6, 'July': 7, 'August': 8,
    'September': 9, 'October': 10, 'November': 11, 'December': 12
}
df_canc['month_num'] = df_canc['arrival_date_month'].map(month_map)
monthly_canc_nights = df_canc.groupby(
    ['month_num', 'arrival_date_month']
)[['stays_in_weekend_nights', 'stays_in_week_nights']].mean().reset_index()
monthly_canc_nights = monthly_canc_nights.sort_values('month_num')

fig_nights = go.Figure()
fig_nights.add_trace(go.Scatter(
    x=monthly_canc_nights['arrival_date_month'],
    y=monthly_canc_nights['stays_in_week_nights'],
    mode='lines+markers', name='Average Week Nights Canceled',
    line=dict(color='#ab63fa', width=3)
))
fig_nights.add_trace(go.Scatter(
    x=monthly_canc_nights['arrival_date_month'],
    y=monthly_canc_nights['stays_in_weekend_nights'],
    mode='lines+markers', name='Average Weekend Nights Canceled',
    line=dict(color='#ffa15a', width=3)
))
fig_nights.update_layout(
    title='Average Nights Lost per Canceled Booking by Month',
    xaxis_title='Month', yaxis_title='Average Number of Nights', hovermode='x unified'
)
fig_nights.show()
```



5. SUGGESTIONS & ACTION PLAN

I. Implement the "PRT Commitment Discount"

- **Targeted Pricing Strategy:** Offer a permanent 12% dynamic discount specifically for domestic reservations in Portugal (PRT) with lead times above 30 days under a strictly non-refundable rate plan.
- **Expected Outcome:** Converts high-volatility domestic placeholders into guaranteed booked nights, decreasing the cancellation rate by an estimated 15-20% in this demographic.

II. Weeknight Optimization & Restructuring of Room Availability

- **Weekday Guardrails:** Since weeknight stays are highly susceptible to cancellation spikes, introduce a mandatory deposit of 1 night's stay for any weekday reservations exceeding 3 nights during high-demand months (June, July, August).
- **Overbooking Algorithm Adjustment:** Calibrate the reservation system's overbooking allowances to reflect higher weekday cancellation rates during summer peaks.

III. Seasonal Marketing Campaigns & Value-Add Structures

- **Off-Peak Dynamic Discounts (Nov - Jan):** Instead of dropping room rates flat (which hurts brand equity), package low-season months with high-perceived-value services (e.g., complimentary spa credits or local dining tours) to boost booking commitments.
- **Early Bird Lock-in:** Implement aggressive booking windows with 20% discount rates during September and October for next summer's travellers, applying a rigid deposit system.

> **Executive Summary & Stakeholder Next Steps**
> By executing these targeted dynamic pricing controls, non-refundable incentives in Portugal, and weekday deposit friction rules, the hotel group can mitigate revenue leakage, stabilize room occupancy, and reduce overall cancellation rates from 37.04% down toward optimal industry benchmarks.